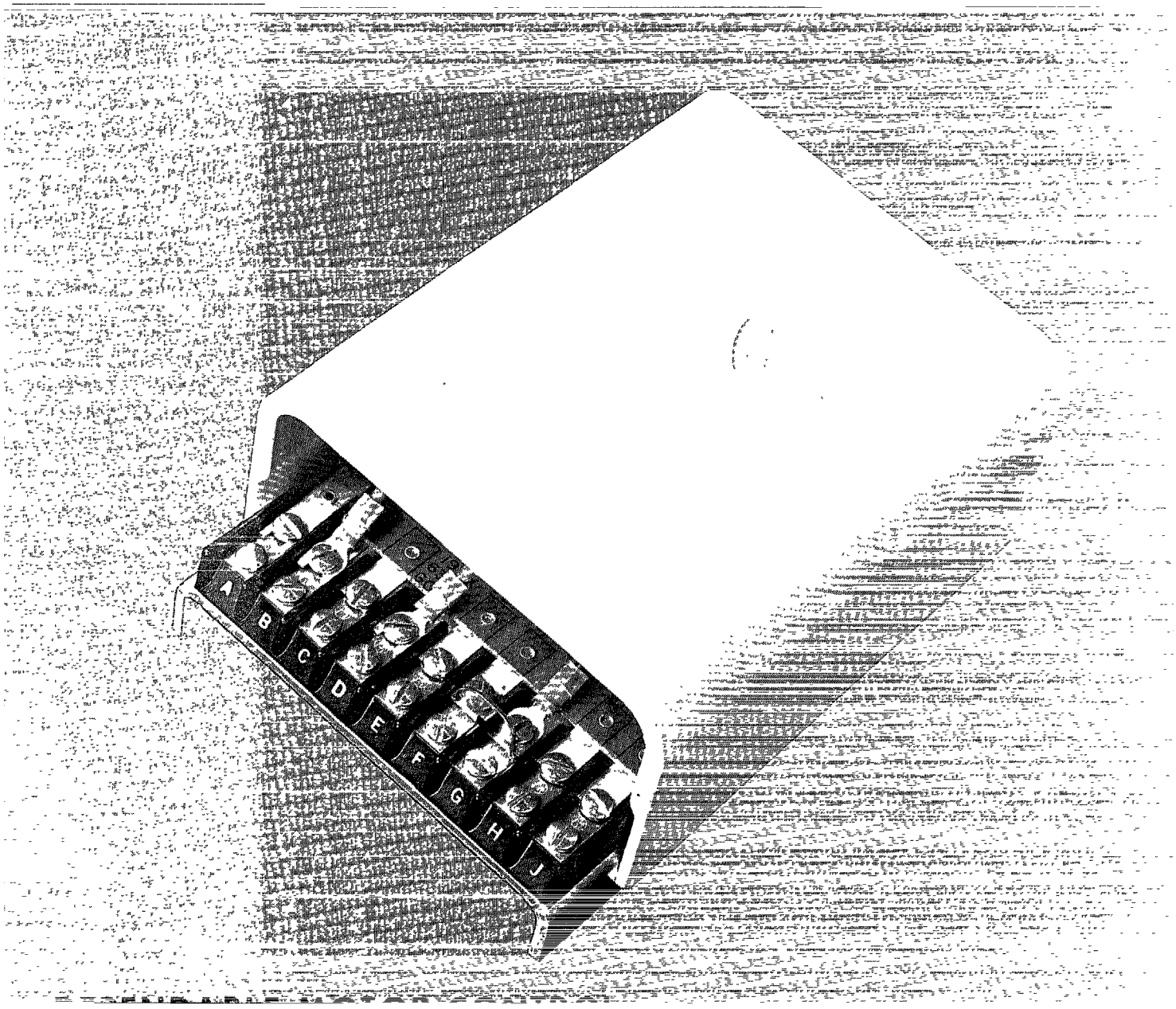
INSTRUCTIONS

GEK-47602

STATOTROL* ADJUSTABLE SPEED DRIVE

331X652AA CONTROL MODULE



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331X652AA
STATOTROL CONTROL MODULE

INTRODUCTION

This manual contains general information on the installation, operation and servicing of General Electric Statotrol Control Modules. When applicable, additional instructions and/or diagrams will be supplied with the equipment to supplement this manual.

The module is used in an electronic adjustable speed drive which controls the speed of fractional and small integral horsepower DC motors. This drive is suitable for a variety of applications in that many modifications are available.

DESCRIPTION

A typical Statotrol Drive will normally include a motor (or gear motor) a controller and an operator's control station.

CONTROL MODULE

The control module converts the alternating current (AC) from the power lines to direct current (DC) to drive the motor. Silicon diodes are used to supply constant field voltage and a silicon controlled rectifier (SCR) to supply adjustable armature voltage. The module is equipped with a special circuit to limit surge current when starting and reversing the motor. Standard fusing and current limiting provide rectifier protection. A standard module will include a maximum speed adjusting potentiometer.

MOTOR

The motor is a DC shunt motor of a design suitable for use with controller furnished. Motor voltage ratings are 75 volt armature with 50 volt field and 140 volt armature with 100 volt field for drives operating on 115 volts AC and 230 volts AC respectively.

RECEIVING AND STORAGE

As soon as the equipment is received it should be unpacked and examined for any damage sustained in transit. If damage is evident, a damage claim should be filed immediately with the transportation company, and the nearest Apparatus Sales Office of the General Electric Company promptly notified.

The equipment should be unpacked carefully to avoid damaging the apparatus. Particular care should be exercised to prevent small parts from being mislaid or thrown away with the packing material.

If the equipment is not to be used as soon as it is unpacked, it should be stored in a clean, dry place and protected against accidental damage. The packing case supplied provides normal storage protection.

INSTALLATION

MOUNTING THE MOTOR

Refer to the drawing furnished with the motor for location of mounting holes and over-all dimensions.

MOUNTING THE CONTROL MODULE

CAUTION

TEMPERATURE AROUND MODULE WHILE IT IS OPERATING MUST NOT EXCEED 40°C (104°F). CONTROL MODULE MUST ALWAYS BE MOUNTED VERTICALLY TO ALLOW MAXIMUM AIR FLOW OVER THE HEAT SINK.

1. Remove plastic cover from control module (see Figure 1).

2. Mount control module using hole locations and over-all dimensions shown on outline supplied.

INSTALLING INTERCONNECTING WIRING

Note that control module contains a fuse in one line only; fusing and disconnects should be installed in accordance with good electrical practice and applicable codes.

Make all connections to the motor at the terminal studs which are covered by a removable plate.

Make connections to the control station by removing the cover and bringing wires to the terminals. If Control Station leads must be run with other power or control wiring, the speed pot leads must be either shielded or run in a separate conduit.

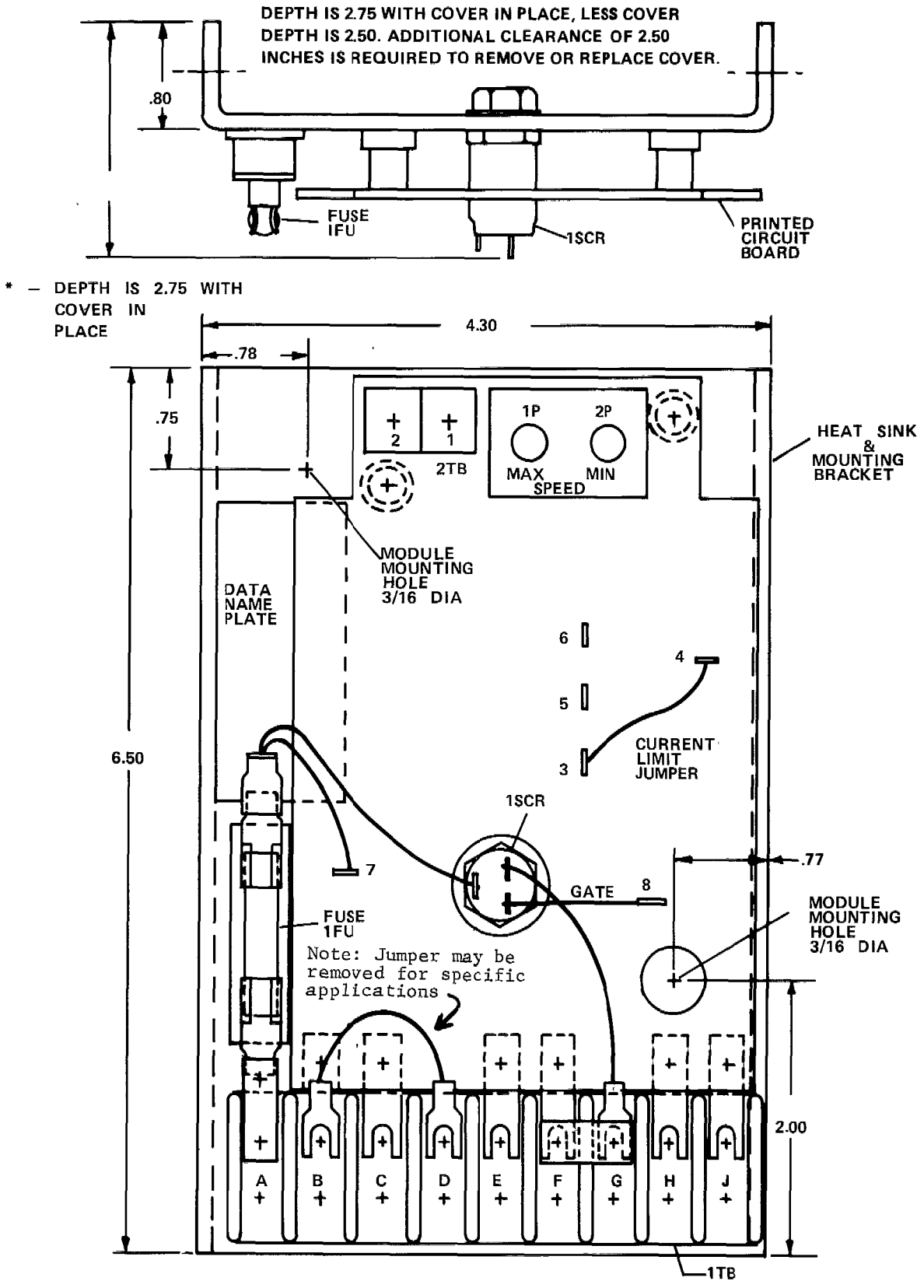
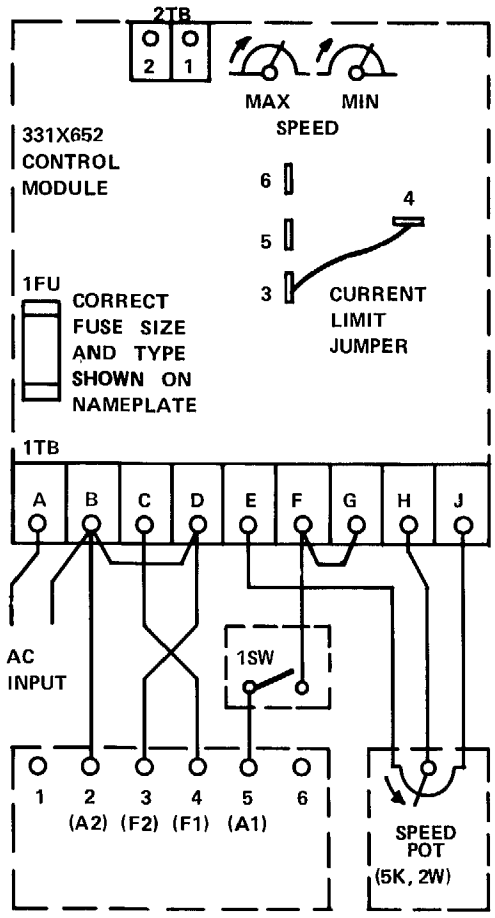


Figure 1 Control Module Assembly & Outline Drawing.



MOTOR ROTATION IS CLOCKWISE. (VIEWED FROM THE COMMUTATOR END) FOR THE CONNECTION SHOWN. FOR OPPOSITE ROTATION REVERSE. THE ARMATURE CONNECTIONS.

Figure 2 Customer connections to 331X652 module.

DIRECTION OF MOTOR ROTATION

If the connections between the motor and the controller are made as shown on the connection diagram, the motor will turn in the clockwise direction when viewed from the commutator end. For opposite rotation of the motor, interchange connections to A_1 and A_2 motor terminals.

The tag on the motor shows its connections.

MAKING PROPER LINE VOLTAGE CONNECTION

This controller operates on 50/60 cycle AC. Input voltage is either 115 volts or 230 volts, not both. Check nameplate to determine proper input voltage. A transformer is required where supply voltage deviates more than ± 10% from the controller rating. IF TRANSFORMER IS REQUIRED, REFER TO DIAGRAM ON THE TRANSFORMER FOR PROPER CONNECTIONS.

CAUTION

DO NOT DEPEND ON CONTROLLER FUSE FOR MOTOR OVERLOAD PROTECTION. THE FUNCTION OF THIS FUSE IS TO PROTECT THE SCR.

The connection diagram (refer to Figure 2) shows how to make connections and the wire size to use for a standard controller.

Make all connections to the controller by bringing wires to the terminal boards.

The input power lead connected to terminal 1TB-B (see diagram) should be grounded whenever possible.

CAUTION

WHILE GROUNDING IS ALWAYS DESIRABLE, IT IS NOT ALWAYS POSSIBLE. THE FEASIBILITY OF THIS ACTION WILL DEPEND ON HOW THE INCOMING POWER SOURCE IS GROUNDED AND THIS MUST BE CONSIDERED CAREFULLY BEFORE POWER IS APPLIED.

OPERATION AND ADJUSTMENT

INITIAL OPERATION

WARNING

HIGH VOLTAGE ELECTRIC SHOCK CAN CAUSE SERIOUS OR FATAL INJURY. BE CAREFUL NOT TO COME IN CONTACT WITH LIVE PARTS.

When all connections have been made correctly, apply power to the drive. Turn the speed potentiometer counterclockwise to its minimum setting. No warmup time is required, so the start button can be pressed immediately. Turn the speed potentiometer clockwise slowly and the motor will gradually gain speed. Turning in the other direction will reduce the speed.

If direction of motor rotation is other than the desired one, change motor connections as described under Installation.

MAXIMUM SPEED ADJUSTMENT

Check the motor speed with the speed potentiometer turned fully clockwise. If the motor speed is higher than

rated, turn the maximum speed adjusting potentiometer (Figure 1) counterclockwise until rated speed is obtained.

MINIMUM SPEED ADJUSTMENT

Check the motor speed with the speed potentiometer turned fully counterclockwise. Turn the minimum speed adjusting potentiometer clockwise until the desired minimum speed is obtained. Note: On some models the minimum speed pot is not connected.

NORMAL OPERATION

The drive can be started with the speed potentiometer set at any desired speed. The motor will start properly and come up to the selected speed. (NOTE: The inertia load reflected to the motor shaft should not exceed the rotor inertia of the motor.) The motor can be stopped when it is running at any speed

Changes in load on the motor up to its rated load will not change the motor speed by any significant amount, because the electronic controller corrects quickly and automatically for changes in load. Changes in line voltage and changes in controller and motor temperature will change the motor speed.

MAINTENANCE

MOTOR

INSPECTION AND LUBRICATION OF THE MOTOR

Make sure that the motor mounting bolts are tight. A loose mounting bolt can cause considerable noise and may result in excessive wear.

Inspect the commutator, the brushes, the brush contact surfaces, and the brush springs. A rough commutator may often be detected by the sound of the motor. Inspect the commutator for color and condition. It should be clean, smooth, and a polished brown color where the brushes ride. A blue color indicates overheating.

Roughness of the commutator should be removed by sand papering or stoning (never use emery cloth or emery stone) For this operation, run the motor without load. If sand paper is used, wrap it partly around a wood block. Press the sand paper or stone against the commutator with moderate pressure while the motor is running and move it back and forth across the commutator surface. It may be necessary to remove the motor from its mounting since the commutator surface is only accessible through the air vent on the underside of the motor casting.

If the commutator is found to be dirty when the motor is inspected, it should be wiped clean with a piece of canvas or other cloth that is free from lint.

Brush inspection is important. The first essential requirement for the satisfactory operation of brushes is free movement of brushes in the holders.

Check the brushes to make sure that they will not wear down too far before the next inspection. When new brushes are installed, they should be permitted to run without load long enough to assure a good fit.

Make sure that each brush surface in contact with the commutator has the polished finish that indicates good contact, and that the polish covers essentially all the brush surface. Check the springs that hold the brushes against the commutator. Improper spring pressure may lead to commutator wear and sparking. The correct spring pressure is 2 to 2-1/2 pounds per square inch of brush in contact with the commutator.

The simplest and most important part of motor maintenance is lubrication. This motor has ball bearings which are packed at the factory with grease sufficient for five years of normal operation. At the end of that time, the bearings should be thoroughly cleaned of the old grease and repacked about 1/3 full with high quality ball bearing grease.

CONTROL MODULE

Look inside the enclosure and be sure it is free of dust, chips, oil or other foreign matter. Be sure to remove the cover of the control module and examine the printed circuit card and mounting bracket (which also acts as a heatsink). They both must be clean to provide reliable operation and maximum heat dissipation.

REPAIR

MOTOR

The motor can be repaired like standard DC motors by any competent motor repair service. If an overload relay trips repeatedly, fuses blow often or the motor shows signs of overheating, check the motor for possible overload. If your drive has an overload relay, check its calibration.

CONTROLLER

The controller is designed so that occasional fuse replacement will usually be the only repairs necessary.

The silicon controlled rectifier (SCR) and silicon diodes should seldom, if ever, need replacing under normal operating conditions. A drive malfunction resulting from controller component failure can be quickly corrected by simply replacing the control module. Before replacement is made, check all connections and components used with the control module. The following check list should help:

TROUBLE SHOOTING AND REPAIR

QUICK CHECK LIST TO LOCALIZE TROUBLE

WARNING

HIGH VOLTAGE ELECTRIC SHOCK CAN CAUSE SERIOUS OR FATAL INJURY. BE CAREFUL NOT TO COME IN CONTACT WITH LIVE PARTS.

IF YOUR MOTOR WILL NOT RUN

1. Be sure all connections are correct and tight.
2. Be sure correct a.c. supply voltage is applied to 1TB terminals A and B.
3. Check the fuse on the control module. If it is blown, replace it. (Important: Fuse must be replaced with one of the same type and rating. Refer to fuses listed in Table III. If it blows again, look for an overload on the motor, a short circuit in the wiring or in the motor, or a shorted SCR or field diodes in the control module.
4. Check the motor relay, if used. Be sure the relay will pick up and contacts are free from dirt and corrosion.
5. Check for reference voltage between 1TB terminals H (positive) and F (negative). If no voltage is present, check the speed control potentiometer, interconnecting wiring and soft start or timed acceleration modifications, if provided.
6. Check for motor field voltage between 1TB terminals G (positive) and C (negative).
7. Replace control module with one having the same "331X" number on the nameplate.

IF YOUR MOTOR RUNS FASTER OR SLOWER THAN NORMAL FOR A GIVEN SETTING OF SPEED CONTROL

1. Check AC supply voltage connected to 1TB terminals A and B to be sure it is normal or does not vary more than ± 10% from normal.
2. Check for possible motor overloads or extreme variations in motor loading.
3. Check the reference voltage applied to 1TB terminals H (positive) and F (negative) to be sure it remains constant at any given speed setting. If it varies, check the speed control potentiometer, interconnecting wiring, and soft start or timed acceleration modifications, if provided. Correct voltage reading will be between zero to 12 VDC depending on speed reference potentiometer setting.
4. Check for correct motor field voltage between 1TB terminals G (positive) and C (negative). It should not vary more than ± 10% from normal. See voltage check list.
5. Replace control module with one having the same "331X" number on the nameplate.

IF THE MOTOR RUNS AT FULL SPEED AND CANNOT BE CONTROLLED BY THE SPEED CONTROL

1. Check all connections for poor connection or short circuit.
2. Check the reference voltage applied to 1TB terminals H (positive) and F (negative). If this voltage remains near 10 volts for zero speed setting, check the speed control potentiometer, interconnecting wiring and soft start or timed acceleration modifications, if provided.
3. Replace control module with one having the same "331X" number on the nameplate.

Voltage Check List

A good control module will give the following voltage reading:

Control Module terminal board		Correct DC Voltage Measurement	
		115 volt	230 volt
Pos (+)	Neg (–)	Control Module	Control Module
1TB-B	1TB-C	51 volts ± 10%	102 volts ± 10%
2TB-1	1TB-F	17 volts ± 15%	17 volts ± 15%
1TB-J	1TB-F	10 volts ± 25%	10 volts ± 25%

REPLACING CONTROL MODULE

1. Disconnect AC power from drive.
2. Remove the interconnecting wires from the control module terminal board. Tag the wires so you will be able to replace them on the proper points.

3. Remove the two mounting bolts located near two opposite corners of the mounting bracket.

4. Mount the new control module and replace the wires on the terminal board.

The replacement control module should have the same catalog number as the one being replaced. Check the "Renewal Parts List" at the back of this instruction book for catalog numbers and ordering information for replacement parts.

REPAIRING CONTROL MODULE

The control module consists of an SCR, printed circuit board, fuse, mounting bracket and associated hardware. Several checks are given below to help determine which part is defective. Defective parts can be obtained by ordering the same catalog number as on the defective part or by referring to the "renewal parts list" at the back of this instruction book. It is recommended that renewal parts be stocked to minimize downtime due to a part failure. In the event that the printed circuit board has failed, replacement of the control module rather than repair of the board is usually the least costly approach.

Before attempting to repair the control module, disconnect and tag all the external wires from the module terminal board, remove the module mounting bolts, and remove the module from its mounting location.

Visually inspect the SCR and printed circuit board for obvious signs of damage. Discoloration on the printed circuit board, cracking or burning of any component can be detected easily and, if present, the module should be replaced.

The SCR can be checked with a volt-ohm meter switched to the X 1K. ohm scale. Measure the resistance between control module terminals F and 7 (other side of fuse from terminal A). The meter should read 100 K. ohm or higher. If the reading is significantly lower, disconnect the wire from tab 7 on the printed circuit board and repeat the measurement. A reading of near infinite ohms is correct. If the reading is significantly lower the SCR should be replaced.

If the SCR checks good, one or more components on the printed circuit card are probably defective. A good card will give the following resistance readings when measured with a voltohmmeter switched to the X 1K scale.

CARD TERMINALS

POS (+)	NEG (–)	RESISTANCE MEASUREMENT
A	C	500K to infinity
B	C	500K to infinity
C	A	1 to 3K
C	B	1 to 3K
A	B	300 to 1200 ohms*
E	F	0 to 3K*
2TB1	F	Above 100K
F	2TB1	2 to 3K
J	2TB1	0 to 6K*
2TB1	8 (stab-out)	500K to infinity*

* same reading with ohmmeter leads reversed.

If any of the above readings cannot be obtained, the control module should be replaced.

Even if both the SCR and printed circuit card appear good based on the above simple checks, it is still possible that they may be defective. Replacement of the control module is the quickest and easiest approach to trouble shooting and repair.

REPLACING THE SCR

1. Disconnect all terminations of the defective SCR from the printed circuit card, fuse block and terminal board, noting where they were connected.

2. Remove the defective SCR from the mounting bracket and carefully clean the area around the SCR mounting hole.

3. Apply silicone grease or an electrical joint compound to the replacement SCR and bracket mounting surfaces.

4. Mount replacement SCR in bracket and tighten the nut (with star washer). The nut should be tightened to 25 In-Lbs to torque (preferably using a torque wrench). Do not overtighten as this may damage the SCR. It will probably be necessary to hold the SCR body hex with a second wrench while tightening the nut.

5. Reconnect the SCR leads to the correct terminals of the module.

PRINCIPLES OF OPERATION

CONTROL MODULE

The control module contains all the power and control circuitry required to operate a motor. The following is a brief description of control module operation.

The Silicon Controlled Rectifier (1SCR in Figures 1 and 3) is gated by a transitorized firing circuit to apply a controlled pulse of power to the motor armature each cycle of the AC line input. Constant excitation of the motor field is obtained by means of diode rectifiers 1D and 2D. Fuse 1FU on the AC input lead at 1TB-A provides short circuit overcurrent protection for the SCR and other components. A metal oxide varistor protects the SCR, Diodes, and Control components from transient overvoltages.

The desired motor speed is achieved by applying a reference voltage between terminal H (+) and terminal E (–) on the control module terminal board 1TB. This reference voltage is normally obtained from an external speed reference potentiometer connected as shown in Figure 2 or 3. Both maximum and minimum speed adjusting potentiometers are provided on the control module to allow setting of a desired motor speed range for full rotation of the speed reference potentiometer. The voltage source for the speed reference potentiometer is a 12 volt zener Diode in the control module. This makes the speed of the motor relatively insensitive to line voltage variations.

The speed reference voltage at 1TB-H is applied to a transistorized summing amplifier. The other inputs to the summing amplifier are a phase shifted component of the AC input voltage and motor armature feedback voltage. The summation of these three voltages causes the firing circuit to generate a gate pulse to the SCR at the proper point in the AC cycle. The resulting power pulse through the SCR to the motor maintains the desired speed as set by the reference potentiometer. Between each power pulse the motor armature current goes to zero. Thus the feedback voltage just prior to firing the SCR for the next power pulse is the true motor CEMF voltage. Since the CEMF voltage is a fairly accurate measure of motor speed, motor load changes will not affect speed nearly as much and speed regulation will be comparable to a drive having IR compensation.

Motor starting and running currents are limited by a current limit circuit which prevents firing the SCR too early in the AC cycle for a given motor speed. The current limit compares motor armature voltage to a second phase shifted component of the AC line voltage and programs the maximum firing advance of the SCR as a function of the motor speed. As the motor starts running, and its speed and armature voltage increase, the voltage feedback to the current limit circuit automatically programs a new, further advanced, SCR firing point such that the current will be limited to approximately the same value independent of speed.

The amount of phase shift of the phase shifted signal determines the firing signal advance limit for any given motor speed. The amount of phase shift is changed by jumper selection on the module printed circuit board. This permits proper current limit settings for motors having different impedances. Motor armature impedance is affected by the base speed of the motor. A low base speed motor has relatively high armature impedance and a voltage pulse which would produce 150% current on its armature would provide significantly higher current in a low impedance (high base speed) motor.

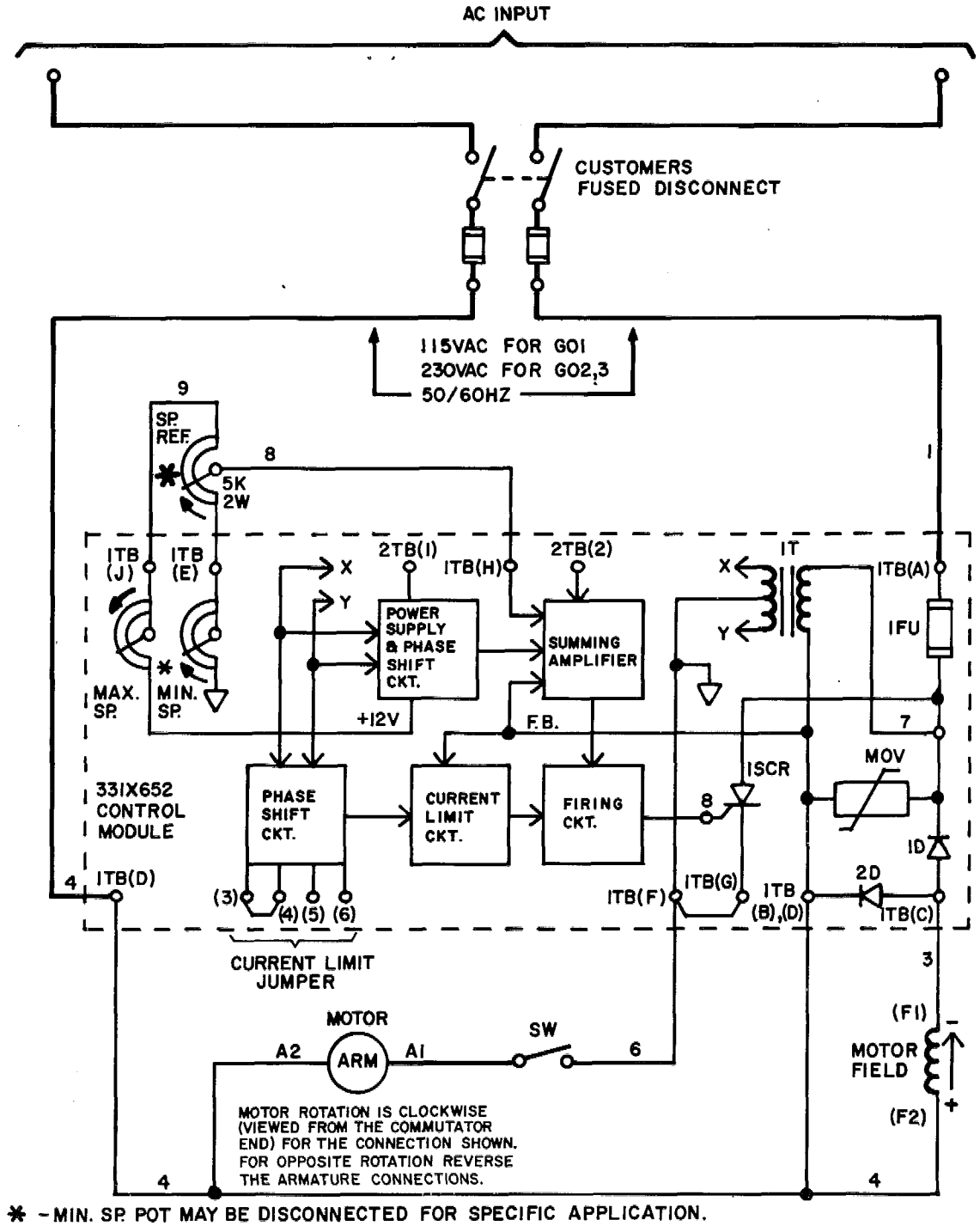


Figure 3 Elementary Diagram of a 331X652AA Statotrol Drive

TABLE I

CONTROL MODULE CROSS REFERENCE LIST			
HP	AC Volts	3SRF Module	Replacement Module
3/4	115	40	331X652AAG01
1-1/2	230	80	331X652AAG02# 331X652AAG03

G02 has MIN SPD pot disconnected

TABLE II

RENEWAL PARTS LIST

FOR 331X652AA STATOTROL CONTROL MODULE

MODEL NO.	QTY. USED	DIAGRAM SYMBOL	General Electric ORDERING NO.	Commercial ORDERING NO. *	DESCRIPTION
331X652AAG01	1	1SCR	942B279BFG01	C147EX32	Silicon Controlled Rectifier
331X652AAG02	1	1SCR	942B279BFG02	C147PX32	Silicon Controlled Rectifier
331X652AAG03	1	1SCR	942B279BFG02	C147PX32	Silicon Controlled Rectifier

* General Electric identification number, do not substitute.

TABLE III

REPLACEMENT FUSE			
MODEL NO.	TYPE	AMPS	PART NUMBER
331X652AAG01	3AB	25	104X109AD056
331X652AAG02	3AB	25	104X109AD056
331X652AAG03	3AB	25	104X109AD056

These fuses must be high interrupting capacity, ceramic body, 250 volt fuses, 1/4 inch in diameter and 1-1/4 inches long. Recommended fuses are Littlefuse 314000 series, type 3AB "Normal D16" fuses and Bussmann type ABC fuses.

WARRANTY

The Contract Warranty for Statotrol drives is stated in the General Electric Apparatus Handbook, Section 98, page 11. The purpose of the following is to provide specific instructions to the Statotrol user regarding warranty administration and how to obtain assistance on out-of-warranty failures.

STATOTROL CONTROLLERS

The warranty covers all major parts of subassemblies of the controller such as the power/control subassembly, modification kits, contactors, etc., but does not provide for replacement on the complete controller.

1. In the event of failure or misapplication during "in warranty", refer to the instruction book to identify the defective part or subassembly.

2. When the defective part has been identified (or for assistance in identification) call:

   STATOTROL DRIVE SERVICE
   Erie, Pennsylvania
   (814) 455-0521

   Before calling, list catalog numbers of the controller, motor, operator's station and any optional modifications. Include orange colored four digit identification number located in lower left corner of controller chassis.

3. Statotrol Service will ship prepaid, at no charge, any parts that have failed in warranty and advise if failed part(s) should be returned to the factory. The factory address is:

   General Electric Company
   Speed Variator Products Section, DCM&GPD
   1100 Lawrence Parkway
   Erie, Pennsylvania 16531
   Attention: STATOTROL DRIVES SERVICE

4. For parts, contact your local General Electric Sales Office or your "Authorized" General Electric Parts Distributor.

   "Exchange" parts at "Exchange" prices may be available for selected items.

STATOTROL MOTORS

Statotrol motor repairs are generally handled by General Electric Small Motor Service Stations or Apparatus Service Shops. For specific instructions on your motor, call STATOTROL DRIVE SERVICE at the number listed above and furnish complete nameplate data.

GENERAL ELECTRIC COMPANY – DIRECT CURRENT MOTOR & GENERATOR PRODUCTS DEPARTMENT
ERIE, PENNSYLVANIA 16531

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